

1. Administrative & Study Identification

Full Study Title: Study of Pembrolizumab (MK-3475) Versus Chemotherapy in Chinese Participants With Stage IV Colorectal Cancer (MK-3475-C66) **Official Title:** A Phase 3 Study of Pembrolizumab (MK-3475) Versus Chemotherapy in Chinese Participants With Stage IV Microsatellite Instability-High (MSI-H) or Mismatch Repair Deficient (dMMR) Colorectal Cancer (MK-3475-C66) **Short Title/Acronym:** MK-3475-C66 **Protocol Number:** MK-3475-C66 **NCT Number:** NCT05239741 **Posting ID:** 685802075566128369 **Sponsor / Company Name:** Merck / Merck Sharp & Dohme LLC **Investigational Product:** Pembrolizumab (MK-3475) / Immune Checkpoint Inhibitor vs. Standard Chemotherapy (e.g., fluorouracil) **Phase of Development:** Phase III **Trial Status:** Recruiting **Disease Areas:** Colorectal Neoplasms **Medical Monitor:** Merck Sharp & Dohme Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of pembrolizumab (MK-3475) compared to standard of care (SOC) chemotherapy in Chinese participants with Stage IV MSI-H or dMMR colorectal cancer. **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), and the safety/tolerability profile of the treatments.

2.2 Background & Rationale To address the need for targeted, highly effective therapies for Stage IV colorectal cancer patients exhibiting the MSI-H/dMMR biomarker. While pembrolizumab has demonstrated superiority over chemotherapy in global populations with advanced MSI-H/dMMR colorectal cancer, this Phase 3 study specifically evaluates its efficacy, safety, and survival benefits as a first-line treatment in the Chinese demographic.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Investigator's choice of 1 of 6 SOC chemotherapy regimens, including combinations with Cetuximab, Bevacizumab, Leucovorin, fluorouracil [ATC: L01BC02], Irinotecan, Oxaliplatin) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 100 participants **Number of Sites:** Multiple clinical sites across China **Trial**

Status: Recruiting (Currently recruiting participants) **Estimated Study Start:** Early 2022 **Estimated Primary Completion:** ~Q4 2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically confirmed diagnosis of colorectal adenocarcinoma that is at Stage IV (per AJCC 8th edition). 2. Centrally confirmed Microsatellite Instability-High/Mismatch Repair Deficient (MSI-H/dMMR) status. 3. Centrally confirmed RAS and BRAF mutation status. 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1 within 3 days before randomization and a life expectancy of at least 3 months. 5. Woman of child-bearing potential (WOCBP) must have a negative highly sensitive pregnancy test (urine within 24 hours or serum within 72 hours) before the first dose. 6. Measurable disease per RECIST 1.1 as assessed by local investigator/radiology. 7. Must provide an archival tumor tissue sample or newly obtained core/excisional biopsy of a tumor lesion not previously irradiated. 8. Well-controlled Hepatitis B (on antiviral therapy for ≥ 4 weeks with undetectable viral load), Hepatitis C (undetectable viral load), or HIV (on antiretroviral therapy).

4.2 Exclusion Criteria 1. Has received prior systemic therapy for Stage IV colorectal cancer (prior adjuvant/neoadjuvant allowed if completed ≥ 6 months prior to randomization). 2. Prior therapy with an anti-PD-1, anti-PD-L1, anti-PD-L2, or agent directed to another stimulatory/co-inhibitory T-cell receptor (e.g., CTLA-4, OX 40, CD137). 3. Severe hypersensitivity ($\geq$ Grade 3) to pembrolizumab and/or any of its excipients. 4. Active autoimmune disease requiring systemic treatment in the past 2 years, or history of (noninfectious) pneumonitis/interstitial lung disease requiring steroids (or current pneumonitis/ILD). 5. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 6. Major operation within 4 weeks of randomization, inadequate recovery, or ongoing surgical complications. 7. Received prior radiotherapy within 2 weeks of start of study intervention. 8. Received a live or live-attenuated vaccine within 30 days before the first dose. 9. Received an investigational agent/device within 4 weeks prior to intervention. 10. Diagnosis of immunodeficiency or receiving chronic systemic steroid therapy (>10 mg daily prednisone equivalent) or immunosuppressive therapy within 7 days prior to first dose. 11. Known additional malignancy progressing or requiring active treatment in the past 5 years (with curative exceptions like basal cell/squamous cell skin carcinoma or in situ cervical/breast cancer). 12. Active infection requiring systemic therapy (e.g., tuberculosis, known viral or bacterial infection) or HIV-infection with a history of Kaposi's sarcoma/Multicentric Castleman's Disease. 13. Had an allogeneic tissue/solid organ transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pembrolizumab (MK-3475) vs. Investigator's choice of SOC chemotherapy (including regimens containing fluorouracil). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or completion of maximum planned treatment cycles.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for symptom management (e.g., anti-emetics). **Prohibited:** Concurrent investigational agents, live vaccines, or systemic immunosuppressive therapies (except as required for immune-related adverse event management).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Informed Consent, Biomarker Confirmation (MSI-H/dMMR/RAS/BRAF), Baseline Imaging, ECOG status
2	0	Randomization, First Dose Administration
3	Q3W / Q9W	Treatment Administration, Safety Labs, Tumor Imaging (Q9W)
4	End of Tx	Final Vital Signs, Exit Interview, Safety Follow-Up (30 Days post-last dose)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints
1. **Progression-Free Survival (PFS) for All Participants:** Defined as time from randomization to first documented disease progression per RECIST 1.1 or death due to any cause, assessed by Blinded Independent Central Review (BICR).
2. **Progression-Free Survival (PFS) for RAS Wild-type Participants:** Assessed per RECIST 1.1 by BICR.

7.2 Secondary Endpoints
1. **Overall Survival (OS) For All Participants**
2. **Overall Survival (OS) For RAS Wild-type Participants**
3. **Objective Response Rate (ORR) For All Participants:** Assessed per RECIST 1.1 by BICR.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine laboratory evaluations, physical exams, and monitoring for immune-related AEs (irAEs).
Serious Adverse Events (SAEs): Must be reported to the sponsor and regulatory authorities within 24 hours.
Data Monitoring Committee (DMC): External, independent Data Monitoring Committee (eDMC) oversees patient safety and efficacy thresholds.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Set for efficacy endpoints; As-Treated/Safety Set for adverse events. **Sample Size Justification:** Target enrollment of 100 participants to adequately power the detection of statistical superiority of pembrolizumab over standard chemotherapy for PFS endpoints. **Handling of Missing Data:** Kaplan-Meier estimates and standard right-censoring rules applied for time-to-event variables (PFS, OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standardized clinical site monitoring (onsite and remote) to ensure strict adherence to protocol, source data verification, and RECIST 1.1 requirements. **Audit Trail:** Secure, CFR 21 Part 11-compliant electronic Data Capture (EDC) system with full audit trails and data clarification forms.

11. Study Identifiers & Governance

Primary ID: MK-3475-C66 **Posting ID:** 685802075566128369 **Registry Number:** NCT05239741 **Sponsor/Lead Organization:** Merck **Study Title:** Study of Pembrolizumab (MK-3475) Versus Chemotherapy in Chinese Participants With Stage IV Colorectal Cancer (MK-3475-C66) **Official Regulatory Title:** A Phase 3 Study of Pembrolizumab (MK-3475) Versus Chemotherapy in Chinese Participants With Stage IV Microsatellite Instability-High (MSI-H) or Mismatch Repair Deficient (dMMR) Colorectal Cancer (MK-3475-C66)

12. Clinical Framework (Colorectal Cancer Specific)

Primary Condition: Colorectal Neoplasms (Stage IV Colorectal Adenocarcinoma) **Diagnosis Threshold:** Central confirmation of MSI-H or dMMR status, as well as RAS and BRAF mutation status. **Medical Methodology:** PD-1 Immune Checkpoint Inhibition vs. Cytotoxic Chemotherapy/Biologics **Optimization Target:** Progression-Free Survival (PFS)

13. Study Procedures & Methodology

Management Pathway: First-line systemic anti-cancer therapy for metastatic disease **Education Protocol (HCP):** RECIST 1.1 evaluation criteria and immune-related adverse event (irAE) early detection and management **Education Protocol (Patient):** Treatment scheduling, symptom reporting, and contraception requirements

(WOCBP negative pregnancy testing requirements) **Quality Audit Mechanism:** Independent central imaging review (BICR) and centralized genomic/biomarker testing

14. Observation & Follow-Up Schedule

Total Duration: Until documented disease progression, unacceptable toxicity, or study withdrawal **Onsite Visit Cadence:** Aligned with treatment cycles **Remote Monitoring Frequency:** Routine pharmacovigilance tracking intervals **Checklist Review Interval:** Tumor imaging assessments regularly scheduled (e.g., every 9 weeks)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				